

Introduction to Pure Mathematics

Problem Sheet 1

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Question 1.1.

- a. Let $u = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $v = \begin{pmatrix} 1 \\ y \end{pmatrix}$ with $y \in \mathbb{R}$ then

$$\|u\| = 1$$

$$\|v\| = \sqrt{1 + y^2}$$

$$\|u + v\| = \sqrt{4 + y^2}$$

Finding $y \in \mathbb{R}$ such that $\|u + v\| = \|u\| + \|v\|$.

$$\sqrt{4 + y^2} = 1 + \sqrt{1 + y^2}$$

Squaring both sides gives us

$$4 + y^2 = 1 + 1 + y^2 + 2 \cdot \sqrt{1 + y^2}$$

Which can be simplified to:

$$1 = \sqrt{1 + y^2}$$

Squaring both sides again leaves us with

$$1 = 1 + y^2$$

Therefore $y = 0$.

- b. let $v_1, v_2, \dots, v_k$ be k arbitrary vectors in $\mathbb{R}^n$. For $k \in \mathbb{Z}_+$ let $P(k)$ be the statement:

$$\|v_1 + v_2 + \dots + v_k\| \leq \|v_1\| + \|v_2\| + \dots + \|v_k\|$$

We will use induction to prove that $\forall k \in \mathbb{Z}_+ P(k)$ is true. For $P(1)$

$$\|v_1\| \leq \|v_1\|$$

is true as the left hand side and the right hand side are equal.

For $P(2)$

$$\|v_1 + v_2\| \leq \|v_1\| + \|v_2\|$$

is true due to the triangle inequality ($\|u + v\| \leq \|u\| + \|v\|$). Now assume that $P(k)$ is true for some integers $k > 2$.

$$\|v_1 + v_2 + \dots + v_k\| \leq \|v_1\| + \|v_2\| + \dots + \|v_k\|$$

Adding $\|v_{k+1}\|$ where $v_{k+1} \in \mathbb{R}^n$ is some arbitrary vector in $\mathbb{R}^n$ to both sides will not change the truth value of the inequality giving us

$$\|v_1 + v_2 + \dots + v_k\| + \|v_{k+1}\| \leq \|v_1\| + \|v_2\| + \dots + \|v_k\| + \|v_{k+1}\|$$

Using the triangle inequality again we can get a second inequality involving the left hand side:

$$\|v_1 + v_2 + \cdots + v_k + v_{k+1}\| \leq \|v_1 + v_2 + \cdots + v_k\| + \|v_{k+1}\| \leq \|v_1\| + \|v_2\| + \cdots + \|v_k\| + \|v_{k+1}\|$$

Due to the transitive properties of inequalities we get that:

$$\|v_1 + v_2 + \cdots + v_k + v_{k+1}\| \leq \|v_1\| + \|v_2\| + \cdots + \|v_k\| + \|v_{k+1}\|$$

Which is the statement $P(k+1)$, so we have shown that if $P(k)$ is true then $P(k+1)$ is true.

So we have shown that for $k > 2$, $P(k) \Rightarrow P(k+1)$ and that $P(1)$ and $P(2)$ are true. Then by the method of induction we have shown that $P(k)$ is true for all $k \in \mathbb{Z}_+$.

An example for which $\|v_1 + v_2 + \cdots + v_k\| = \|v_1\| + \|v_2\| + \cdots + \|v_k\|$ is

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ -4 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 2 \\ 2 \\ -8 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 3 \\ 3 \\ -12 \end{pmatrix}.$$

$$\text{Where } \|v_1\| = 3\sqrt{2}, \|v_2\| = 6\sqrt{2}, \|v_3\| = 9\sqrt{2}, v_1 + v_2 + v_3 = \begin{pmatrix} 6 \\ 6 \\ -24 \end{pmatrix},$$

$$\|v_1 + v_2 + v_3\| = 18\sqrt{2} \text{ and } \|v_1\| + \|v_2\| + \|v_3\| = 18\sqrt{2} = \|v_1 + v_2 + v_3\|$$

However this equality holds for any set of vectors if all the vectors are parallel in the same direction ($\forall i, j \in [1, k], \exists \lambda \in \mathbb{R}_+$ such that $v_i = \lambda v_j$).